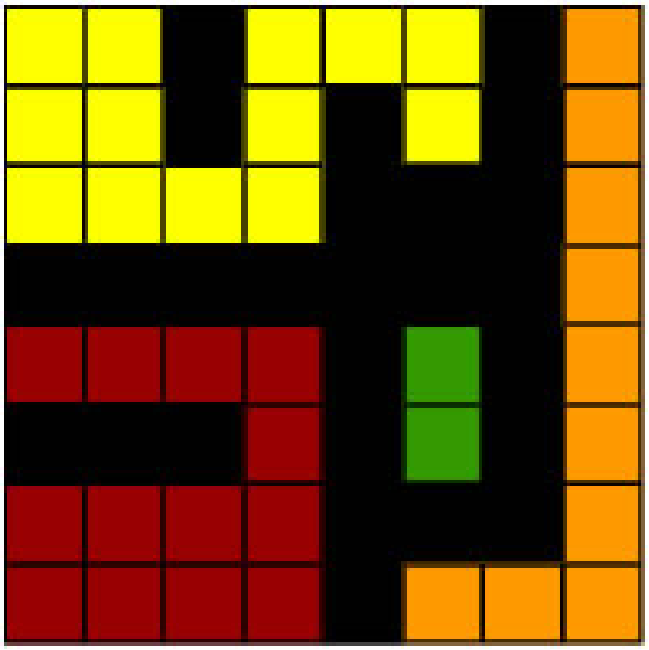
Connected Component Labelling

Image Processing

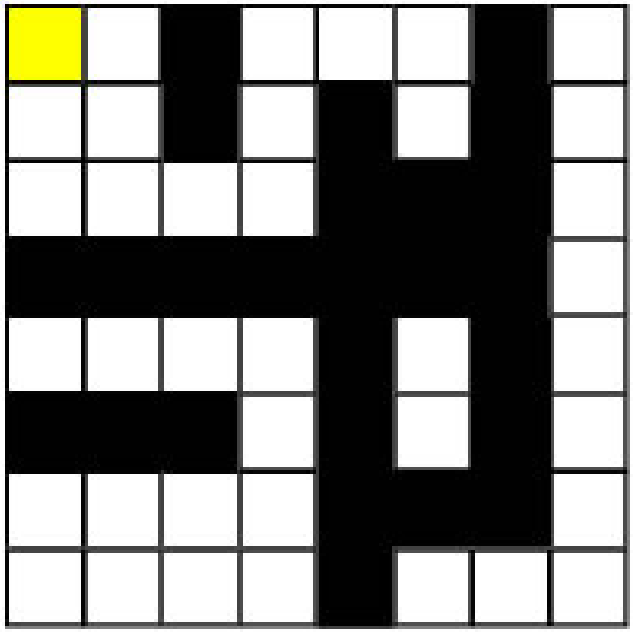
The "blocks" represent a pixel. White means its a '1' and black means its a '0'. We're interested in labeling the non-zero pixels. Ideally, you can just loop at the image and come up with a possible labeling scheme:

1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	1
1	1	1	1	0	0	0	1
0	0	0	0	0	0	0	1
1	1	1	1	0	1	0	1
0	0	0	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	1	1	0	1	1	1

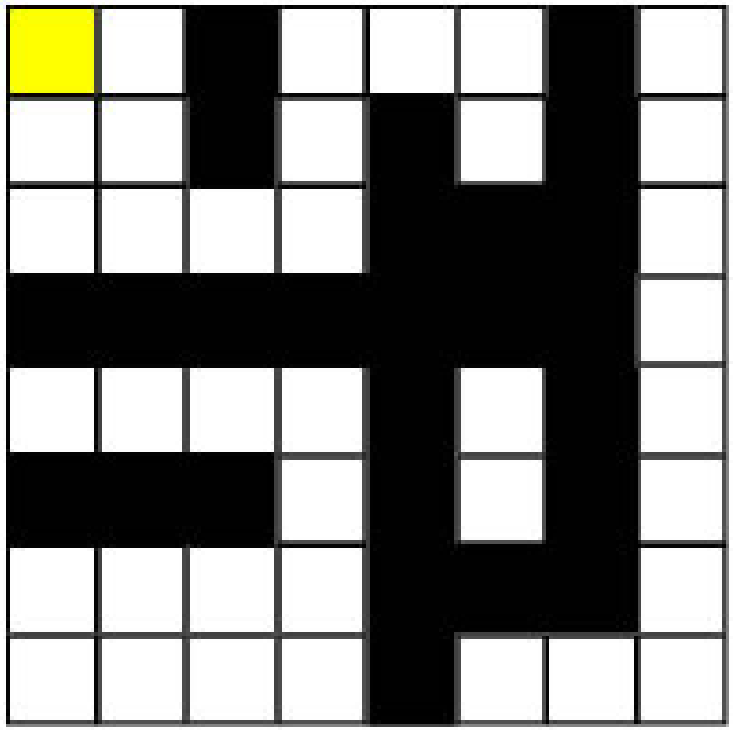


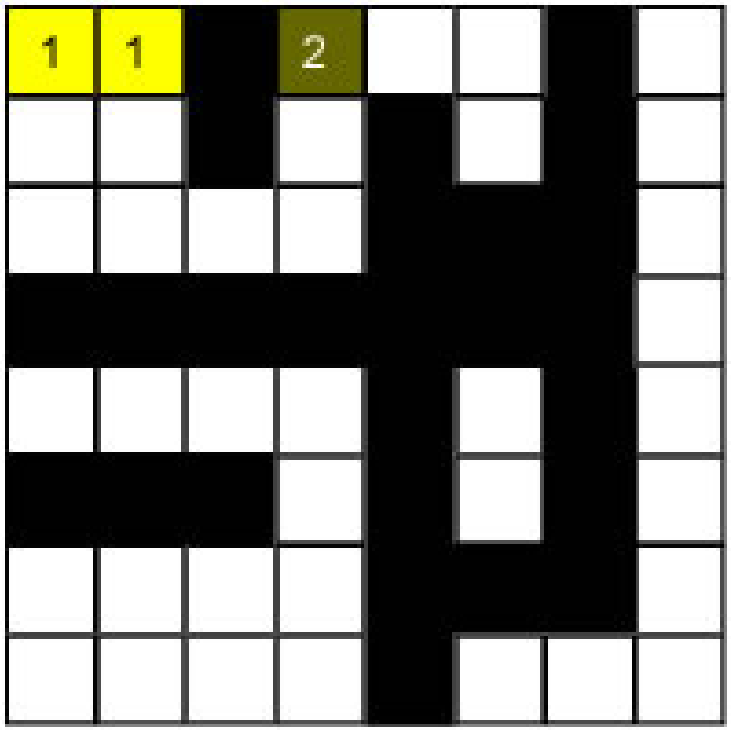
The first pass

1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	1
1	1	1	1	0	0	0	1
0	0	0	0	0	0	0	1
1	1	1	1	0	1	0	1
0	0	0	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	1	1	0	1	1	1

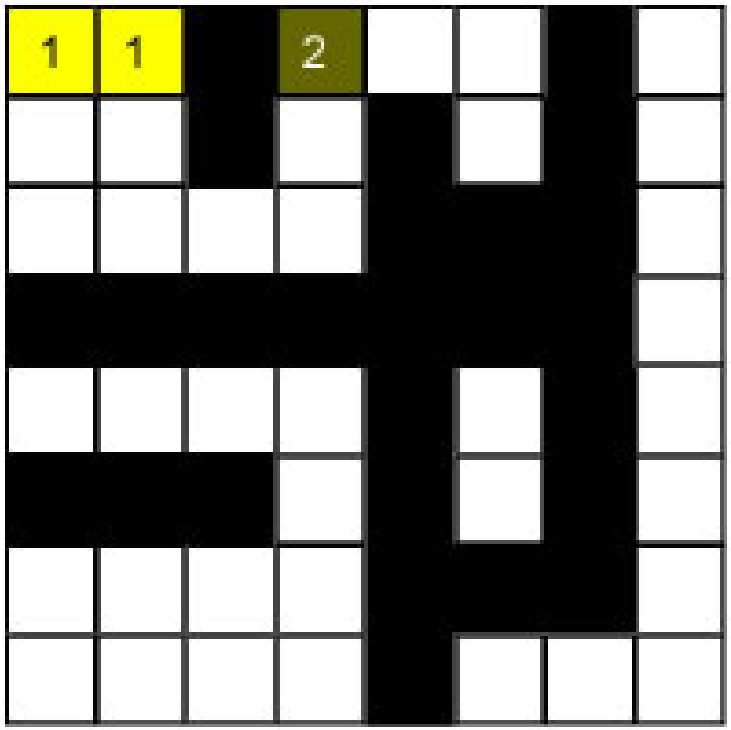


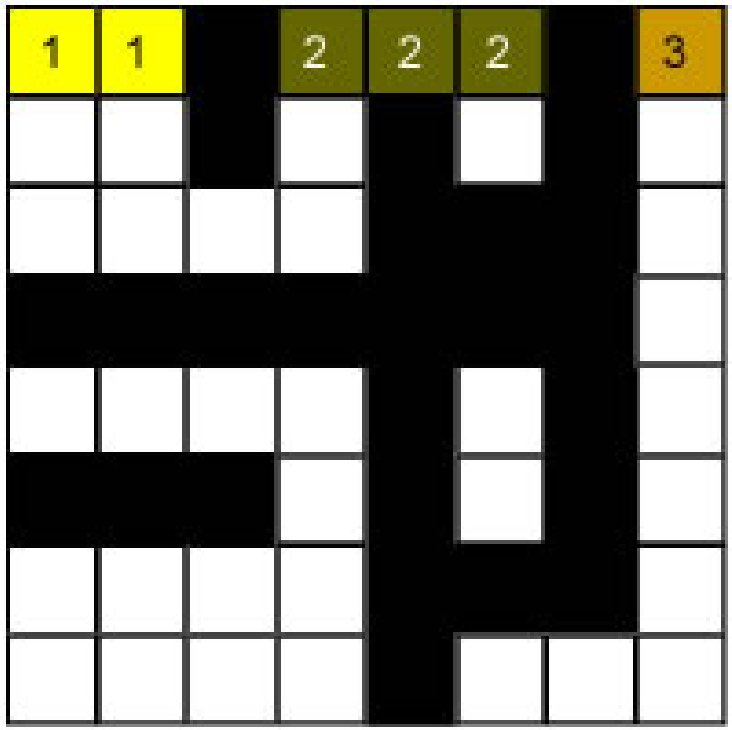
The first pass



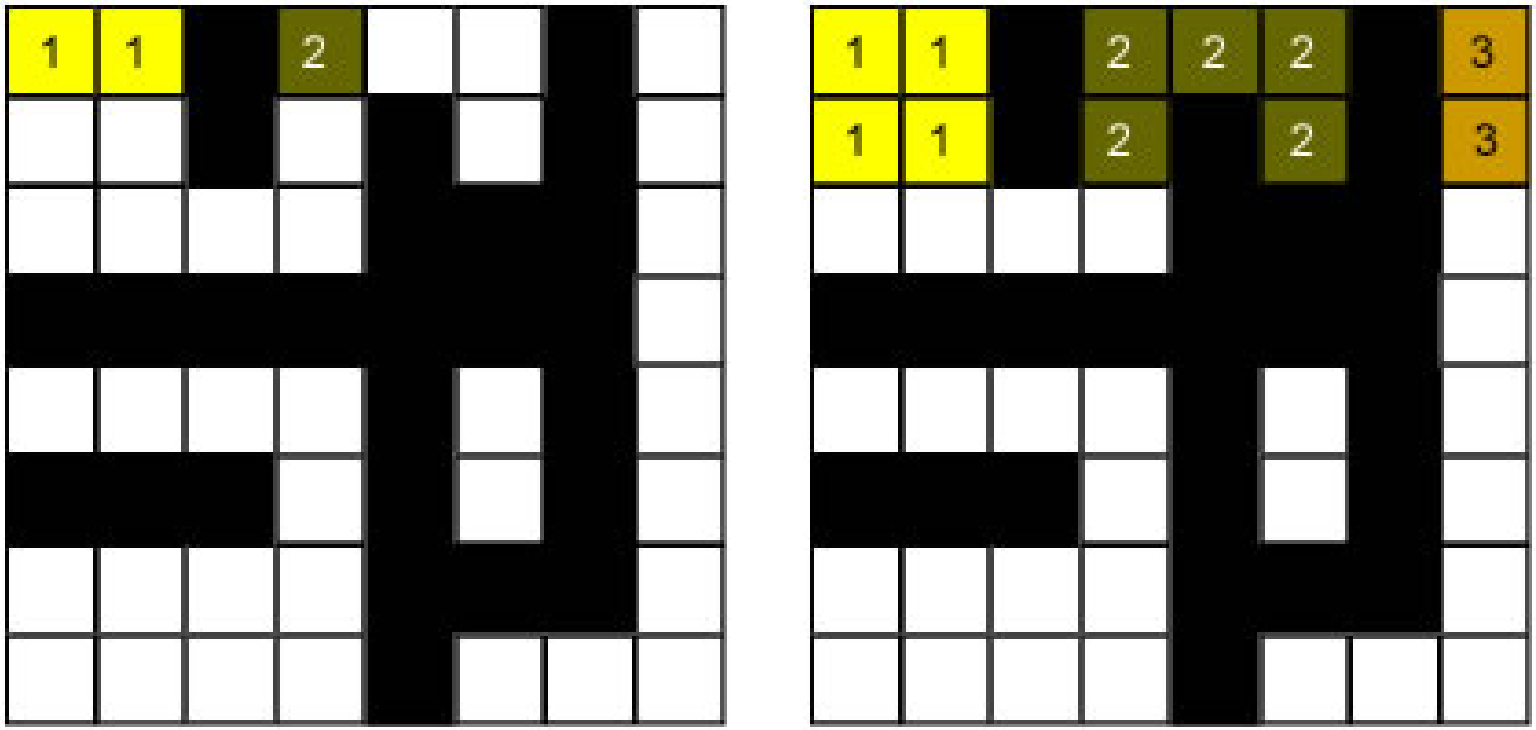


The first pass

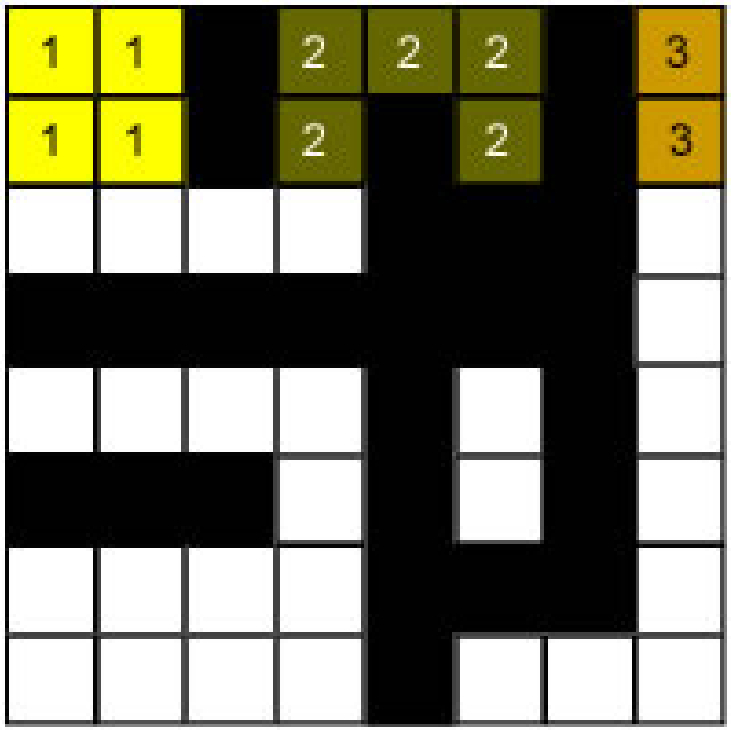


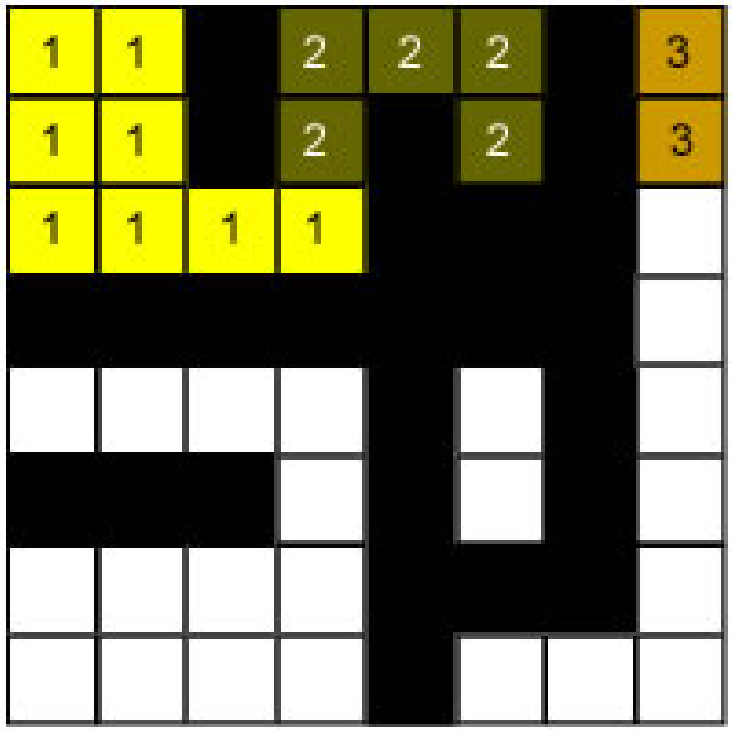


The first pass : Completion of row 2

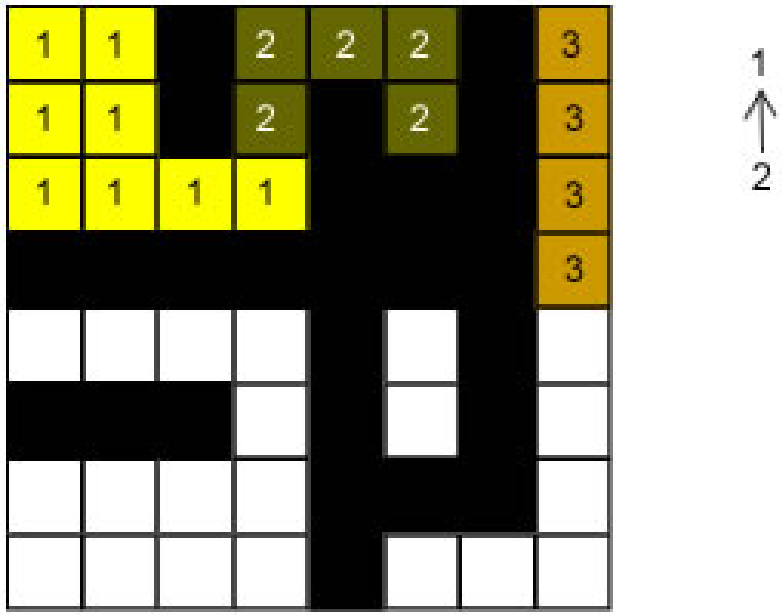


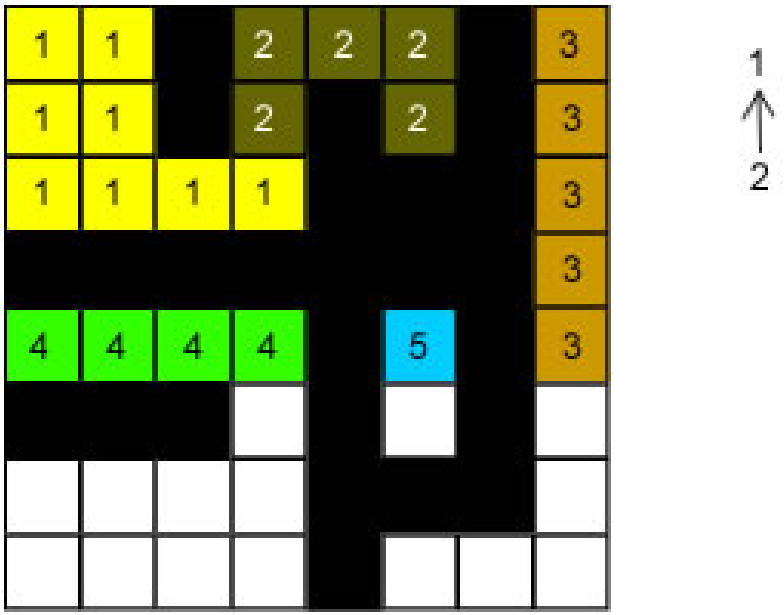
The first pass : Completion of row 2

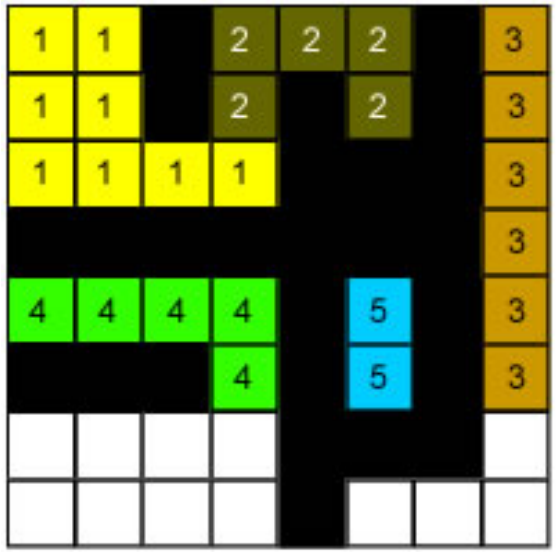




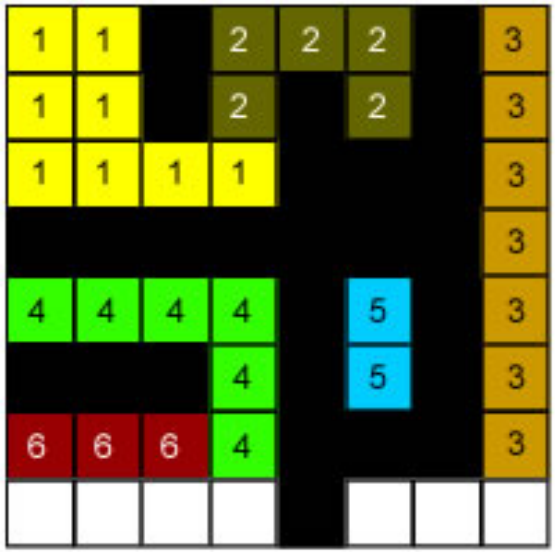



1 1 2 2 2 3
1 1 2 2 3
1 1 1 1 3
3
1
2


1 1 2 2 2 3
1 1 2 2 3
1 1 1 1 3
3
4 4 4 4 5 3
1
2

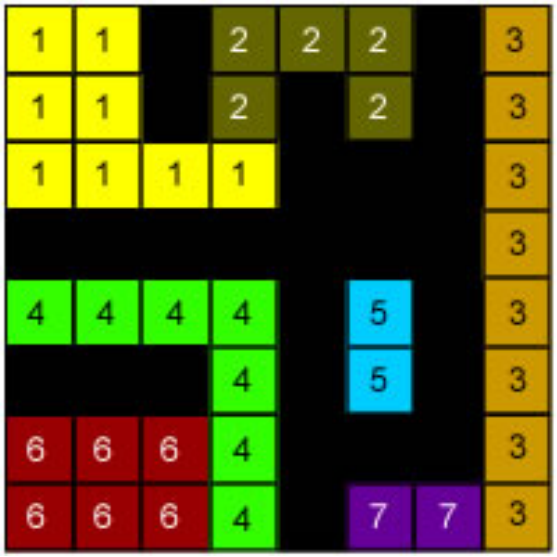


1
2



1
2

4
6

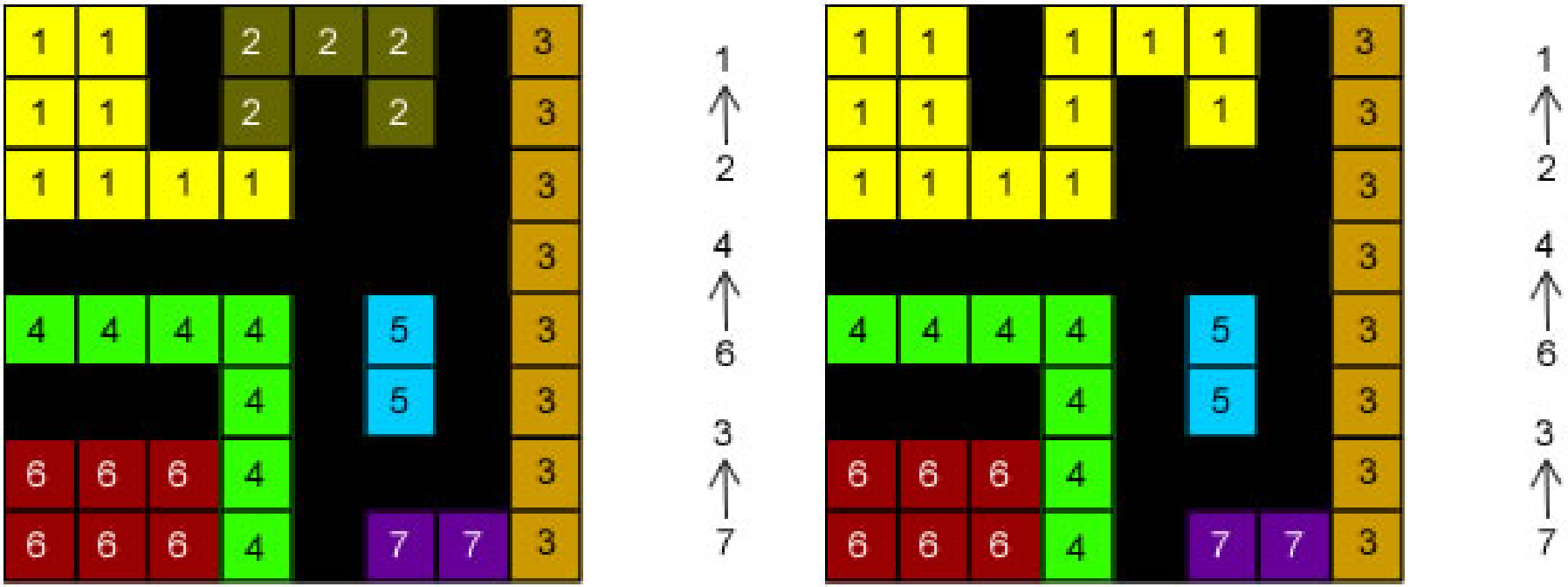


1
2

4
6

3
7

The second pass



The second pass

